



Machine learning classification of EEG responses to pain-related vs non-pain-related stimulus in preterm infants

Lojain Hamwi^a, Hang Du^b, Sara Jasim^a, Xiaogang Wang^a, Vibhuti Shah^c, Carol Cheng^c, Lorenzo Fabrizi^d, Maria Fitzgerald^d, Judith Meek^d, Nicole Racine^e, Ian Stedman^a, Rebecca Pillai Riddell^{a,*}

Abstract

Introduction: Unmanaged pain in preterm infants can lead to long-term developmental consequences. Current pain assessment methods lack specificity, resulting in possible pain mismanagement in Neonatal Intensive Care Units (NICUs). This study explores the application of machine learning (ML) to differentiate between pain-related and non-pain-related cortical activity in preterm infants.

Objective: To evaluate the performance of ML models in distinguishing cortical EEG activity during a painful procedure in preterm infants across different postmenstrual ages (PMAs).

Methods: This observational study was conducted from June 2015 to May 2024 at Mount Sinai Hospital in Toronto, Canada, and University College London Hospital, United Kingdom. EEG data were collected from 72 preterm infants (27 females) during routine heel lance procedures while held in skin-to-skin contact. Infants' gestational ages ranged from 24 to 36 weeks with a mean PMA of 32.87 weeks. Five ML models—XGBoost, support vector machines, Random Forest, Logistic Regression (LR), and convolutional neural networks—distinguished EEG activity pre-heel and post-heel lance.

Results: Model performance was assessed using accuracy and area under the receiver operating characteristic curve (AUC). In the oldest PMA group (≥ 34 weeks), LR achieved the highest mean accuracy (82%) and AUC (0.90). Similarly, LR achieved the highest mean accuracy (70%) and AUC (0.94) in the middle PMA group (32–33 weeks, 6 days). In the youngest group (< 32 weeks), all models except XGBoost performed relatively the same with a mean accuracy of 76% or 77% and a mean AUC of 0.82 or 0.80.

Conclusion: Machine learning models demonstrate potential in distinguishing pain-related cortical activity, offering a pathway for improved neonatal pain assessment in NICUs.

Keywords: Preterm infants, Infant pain assessment, Brain activity, Machine learning, Neonatal Intensive Care Unit

1. Introduction

In 2020, approximately 13.4 million newborns were born prematurely globally.³⁹ Preterm infants often require prolonged care in Neonatal Intensive Care Units (NICUs), undergoing an average of 12 to 17 painful procedures daily in their first 2 weeks of life.^{8,19} Frequent pain exposure can lead to structural and

functional cortical changes, correlating with poorer neurodevelopmental outcomes.^{6,11,14,37,62,63}

Effective pain assessment is essential to mitigate consequences of extended NICU stays.²⁵ However, assessing pain in hospitalized infants is challenging due to their inability to verbally express their pain.⁵² Current pain assessment tools rely on

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

^a York University, Toronto, ON, Canada, ^b Michigan Technological University, Houghton, MI, USA, ^c Department of Paediatrics, Sinai Health System—Mount Sinai Hospital, Toronto, ON, Canada, ^d Department of Neuroscience, Physiology and Pharmacology, University College London, London, United Kingdom, ^e University of Ottawa, Ottawa, ON, Canada

*Corresponding author. Address: Department of Psychology, York University, Toronto, ON, M3J 1P3, Canada. Tel.: (416) 736-2100 ext. 20177; fax: (416) 736-5814. E-mail address: rpr@yorku.ca (R. Pillai Riddell).

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.painreports.com).

Copyright © 2025 The Author(s). Published by Wolters Kluwer Health, Inc. on behalf of The International Association for the Study of Pain. This is an open access article distributed under the Creative Commons Attribution-NoDerivatives License 4.0 (CC BY-ND) which allows for redistribution, commercial and non-commercial, as long as it is passed along unchanged and in whole, with credit to the author.

PR9 10 (2025) e1332

<http://dx.doi.org/10.1097/PR9.0000000000001332>

behavioral (eg, facial expressions, crying) and physiological responses (eg, heart rate, cortisol levels).^{18,47,57} Despite their utility, these methods face challenges in preterm infants, including nonspecificity to pain-related distress, staffing constraints, and insufficient training.^{5,45} The need for improved pain assessment has driven research into artificial intelligence methodologies, including machine learning, to enhance the accuracy and feasibility of infant pain assessment for hospitalized infants.

1.1. Machine learning in infant pain assessment

The application of machine learning (ML) in pain medicine is expanding rapidly.^{13,32,65} In recent years, ML studies have shown promising results in predicting pain intensity, identifying pain patterns, and classifying pain.^{22,30,46} Particularly, machine learning methods have shown high accuracy in detecting facial expressions related to a noxious stimulus.^{7,20,28} However, preterm infants often have their face obscured by medical devices and pain facial behaviours are subtle. Some multimodal measures have combined behavioural and physiological indicators to create a more comprehensive assessment of pain.^{50,63,64} However, these indicators lack specificity to the infant pain response.⁹ Thus, there remains a strong need to bring cortical responses for infant pain outside the laboratory and into clinical practice, as they have the potential to be a pain-specific clinical indicator.^{4,53,60}

1.2. EEG in machine learning

EEG, or electroencephalography, is a noninvasive tool that measures the brain's electrical activity. Automatic analysis of EEGs holds great promise to be feasible and effective, especially for preterm medical care.^{27,51,56} EEG-based algorithms have demonstrated satisfactory results in detecting seizures, sleep staging, and predicting neurodevelopmental outcomes.^{1,2,15,16,29,34,35,38,40,42–44,48}

To date, 2 EEG-based pain assessment machine learning algorithms exist for neonates.^{58,59} These studies included data from both preterm and full-term infants to discriminate brain activity in response to a heel lance vs a control sham. The 2019 study achieved 64% accuracy using EEG-only data, and the 2022 study reached 74% accuracy using data from only preterm infants. However, a significant limitation of their approach was the reliance on a single EEG channel to capture cortical activity. Assessing noxious-evoked brain activity using only peak-to-peak magnitude across a single EEG channel may overlook activity from other critical brain regions and may have reduced accuracy. In addition, the process required human assessment of peaks and excluded artifact-affected EEG traces, limiting ecological feasibility. Finally, subanalyses across different postmenstrual age (PMA) groups were limited by small sample sizes in some subgroups (eg, $n = 3$ in the youngest group).

These initial attempts provide several areas for future development. First, there is a need to incorporate multiple EEG channels and features for a more comprehensive analysis of pain-related cortical activity. Second, methodologies that handle EEG artifacts and use automated processes from the outset are essential for enhancing clinical relevance. Our study aimed to address these gaps by pursuing algorithms that use multiple EEG channels to maximize data and move toward a real-time, continuous pain monitoring system in the NICU.^{3,10,33}

2. Method

The objective of this study was to explore machine learning techniques to distinguish the EEG activity recorded in the 1-second epoch after routine heel lance (heel being held and lancet deployed) from the one-second epoch that preceded the heel lance (no skin-breaking stimulation but heel being held and lancet resting on heel) in preterm infants. As such, an observational study was conducted involving clinically required lances in hospitalized preterm newborns. Data collection took place at 2 tertiary care NICUs: Mount Sinai Hospital (Toronto, Canada) and University College London Hospital (London, United Kingdom) between June 2015 and May 2024. Ethical approval for the study was obtained from the ethics review boards of all institutions involved (York University Human Participants Review Committee Approval Number 2020-034; Mount Sinai Hospital Institutional Review Board Approval 19-02520AMSH; University College London Hospital Ethics Review Board Approval Number IRAS 60231).

2.1. Participants

The data set comprises of 72 mother–infant dyads (all caregivers identified as female in demographic questionnaires that allowed a broad range of gender identities). Preterm infants (27 females) born between 24- and 36-week gestational age (see **Table 1** for demographic information) were recruited only after they were deemed medically stable. Exclusion criteria included congenital malformations, administration of analgesics or sedatives at the time of the study, evidence of perinatal hypoxia/ischemia, hormonal imbalances, or any cardiac or neurological abnormalities. In addition, to help ensure a non-pain-related distress state pre-lance, infants with severe diaper rash or excoriated buttocks were not included.

2.2. Procedure

The timing of the study was always standardized around the infants' scheduled feed time, leading up to their routine blood draw (**Fig. 1**). The data collection process was standardized between sites and the same trained senior neonatal research nurse at each site (who met and synchronized procedures in person) was present for every study. EEG and video recordings were collected continuously throughout the session. First, EEG leads were placed on the infants' scalp while in the cot or incubator. A small amount of EEG preparation gel (Nuprep; D.O. Weaver and Co., Aurora, CO) was gently rubbed onto the infant's scalp, followed by the application of conductive paste (Nihon; Kohden Elefix, Tokyo, Japan) to each electrode to ensure optimal contact. A soft bonnet was then placed over the infant's head to keep the electrodes securely in place. Next, the diaper-clad infant was transferred from the cot to the mother, who settled into a comfortable seated position for approximately one hour of skin-to-skin contact with the child against her bare chest. This position was maintained consistently across all procedures.

Administration of sucrose prior to the heel lance was site-dependent; infants in Canada ($n = 45$) received 24% sucrose shortly before the procedure, whereas infants in UK ($n = 27$) did not receive sucrose. The amount of sucrose given to Canadian infants was dependent on their weight: 0.2 mL of 24% sucrose was given for infants weighing less than 1500g; 0.5 mL for infants weighing 1500–2500g; and 1 mL for over 2500g. At the end of the resting period, a research nurse performed the routine heel lance. As a preparatory step, the lancet was positioned against the

Table 1
Demographic data of participants.

	Site		
	UCLH, UK (<i>n</i> = 27)	Mount Sinai, Canada (<i>n</i> = 45)	Combined sample (<i>n</i> = 72)
Gestational age at birth (wk)			
Mean (SD)	31.37 (3.88)	29.35 (1.39)	30.11 (2.77)
Median (IQR)	32 (7.07)	29.14 (1.86)	29.29 (3.11)
Range	24–36.86	27–32.43	24–36.86
Postmenstrual age at the time of the study (wk)			
Mean (SD)	33.76 (2.72)	32.34 (1.75)	32.87 (2.26)
Median (IQR)	34.29 (5.58)	32.29 (2.86)	32.57 (3.47)
Range	29.86–38	28.71–36.29	28.71–38
Postnatal age at the time of the study (d)			
Mean (SD)	14.89 (15.51)	21.18 (8.83)	19.13 (12.01)
Median (IQR)	11 (13.50)	19 (10.50)	17 (13.50)
Range	0–61	7–41	0–61
Sex			
Female	12	15	27
Male	15	30	45
Birth weight (g) [mean (SD)]	1700.55 (790.2)	1168.85 (213.07)	1349.08 (546.57)
Apgar score (5 min) [mean (SD)]	8.75 (1.77)	8.42 (1.15)	8.53 (1.39)

The Apgar score at 5 minutes evaluates a newborn's condition based on 5 criteria: Appearance (skin color), Pulse (heart rate), Grimace response (reflex irritability), Activity (muscle tone), and Respiration (breathing rate and effort). Each criterion is scored between 0 and 2, with a total score ranging from 0 to 10. A score of 7 to 10 indicates good health. IQR, interquartile range; SD, standard deviation.

infant's heel for 30 seconds as a non-noxious stimulus before the heel lance broke the skin. The exact moment of skin break was synchronized with the ongoing EEG recording using an accelerometer mounted on the lancet to detect the blade's release.

2.3. Data processing

2.3.1. Signal acquisition

Data from 10 electrodes (F7, F8, T7, T8, P7, P8, O1, O2, Cz, Pz) were recorded according to a subset of the international 10/20 electrode placement system (eFig. 1, <http://links.lww.com/PR9/A340> in Supplement). EEG signals collected in the United Kingdom were sampled at a rate of 2000 Hz and recorded using a Compumedics Neuroscan SynAmp2 system, then down-sampled. Canadian data were sampled at a rate of 256 Hz and recorded using Moberg Component Neuromonitoring System Envision. The recording reference was placed at Fz. The onset of the stimulus was labeled as “1” (heel lance) and whereas the other epoch was labelled as “0” (control).

2.3.2. Preprocessing

EEG data were preprocessed using the Harvard Automated Processing Pipeline for Electroencephalography *HAPPILEE* pipeline²⁶ (see pipeline in eFig. 2, <http://links.lww.com/PR9/A340> in Supplementary Materials) in MATLAB (*R2023b*).³¹ Raw data were filtered with a notch filter (48–52 Hz for data from the United Kingdom; 58–62 Hz for Canadian data) to remove power line noise. Wavelet thresholding with a soft empirical Bayesian level-dependent threshold was applied using *coiflet* wavelets optimized for EEG data.²⁶ Data from the United Kingdom was down-sampled to 250 Hz to match the sampling rate of the Canadian data. All EEG data were re-referenced to the common average. To reflect real-world NICU conditions, where raw EEG signals often contain artifacts, only basic artifact removal was performed autonomously using *HAPPILEE*. Artifacts present in long-term EEG recordings were incorporated into the data-driven training process, an approach recognized as effective in developing ML algorithms for similar applications.⁴⁰

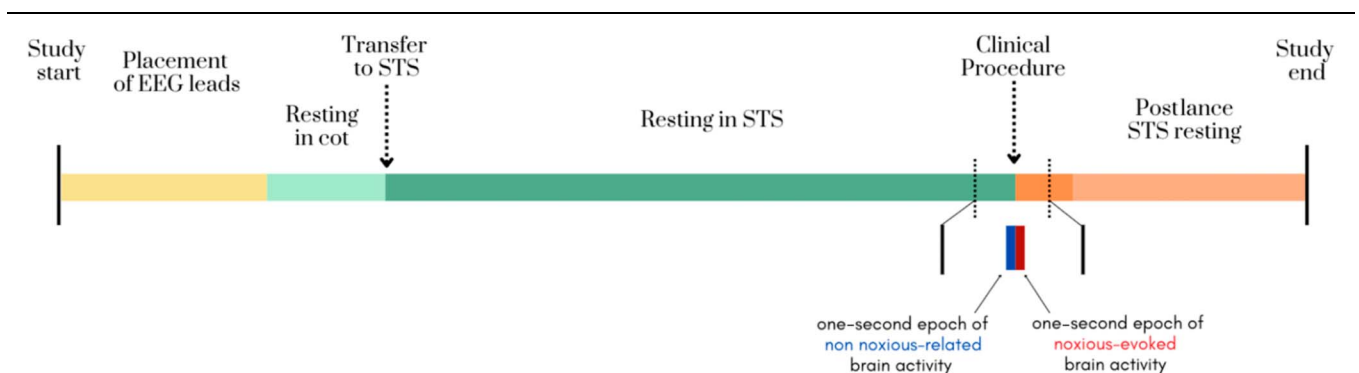


Figure 1. Overview of the study procedure. The study involves EEG placement, transfer to skin-to-skin contact (STS), clinical procedure (heel lance), and postlance STS resting. The blue segment represents 1 second of non-noxious-related brain activity, while the red segment indicates 1 second of noxious-evoked brain activity immediately post-lance. Estimated durations for each study segment are as follows: placement of EEG leads (45 minutes to 1 hour), resting in cot (10–20 minutes), resting in STS (45 minutes to 1 hour), and postlance STS resting (10–20 minutes).

2.3.3. Feature extraction

A comprehensive set of frequently used features in neonatal EEG analysis was compiled using the NEURAL software package⁴¹ (v0.3.1) within the MATLAB environment (Math-Works Inc). The package generates 30 features grouped into 4 categories (see eTable 1, <http://links.lww.com/PR9/A340> in Supplementary Materials): amplitude, spectral, connectivity, and interburst interval. Features were calculated at 4 frequency bands (FB1: 0.5–3 Hz; FB2: 3–8 Hz; FB3: 8–15 Hz; and FB4: 15–30 Hz) relevant to the developmental stage of the infants in the sample. For each infant, 2 one-second epochs were extracted—the 1 second immediately preceding the heel lance and the one second immediately after the heel lance—to create a balanced data set. This choice is supported by prior work demonstrating that noxious-evoked cortical responses in neonates typically occur within a well-defined time window (~400–700 ms) after stimulus onset.^{17,23,49} Given the short duration of these epochs, certain features, such as connectivity and interburst interval, were not considered accurate and were disregarded.

Sucrose administration status (yes/no) was merged into the feature table alongside the EEG features. This covariate was carried through the entire feature-selection pipeline and tested for retention after correlation filtering, ANOVA, and XGBoost importance ranking.

Finally, to ensure full reproducibility, a fixed random seed was applied to every stochastic step (train/test split, feature-selection filtering, and ten-fold grid-search cross-validation) so that identical results can be obtained on each run.

2.3.4. Feature selection

To reduce multicollinearity, Pearson correlation coefficients were calculated between features. For highly correlated pairs ($|r| > 0.8$), the feature showing the greater difference between the pre-lance vs post-lance classes (assessed via *t*-tests or Mann–Whitney *U* tests) was retained. Those surviving this filter entered an ANOVA ($P < 0.05$) and were then ranked by weight-based importance in a XGBoost classifier. This tree-based ensemble method captures nonlinear effects and outputs a direct measure of each feature's contribution to model accuracy, making it a robust way to identify which EEG features best discriminate pre-lance vs post-lance epochs. The final feature set was normalized with a standard scaler before modeling. Classification analyses were implemented using custom scripts and libraries, ie, NumPy, SciPy, and Scikit-learn.

2.3.5. Classification

Five algorithms were evaluated for binary classification of the pre-lance (non-pain stimulus) vs post-lance (pain-related stimulus) epochs: support vector machines (SVM), random forests (RF), logistic regressions (LR), convolutional neural networks (CNN), and gradient boosting (XGBoost).

2.3.6. Age stratification

Given the rapid brain development in infants, model performances were explored by splitting the data into 3 data sets based on PMA at the time of the study: Youngest group: PMA ≤ 31 weeks, 6 days ($n = 27$); Middle group: PMA 32 weeks, 0 days to 33 weeks, 6 days ($n = 20$); and Oldest group: PMA ≥ 34 weeks ($n = 25$). This stratification was important because the infant brain's structural and functional properties change exponentially

early in development, and different stages of maturation might exhibit distinct EEG patterns in response to pain.³⁶

2.3.7. Performance evaluation

Performance was assessed entirely at the *participant* level to avoid any possibility of data leakage. The sample was partitioned into 5 non-overlapping folds. In each iteration, 20% of infants were held out as a test set while the remaining 80% formed the outer training set. Within each outer training split, a ten-fold GroupKFold cross-validation was used to optimize XGBoost hyperparameters. Hyperparameter tuning was focused on XGBoost because its performance is particularly sensitive to parameter choices, whereas other classifiers (SVM, RF, LR, and CNN) could achieve optimal results using default settings and thus were simply retrained on the full 80% outer training data. Each model's accuracy, precision, recall, specificity, F1-score, and area under the receiver operating characteristic curve (AUC) (see eTable 2, <http://links.lww.com/PR9/A340> in the Supplementary Materials for definitions) were measured on the held-out infants. This process was repeated for all 5 outer folds, and the results are reported as the mean ± SD of each metric across those independent test sets to summarize each model's ability to discriminate between classes.

3. Results

The most influential features identified by the XGBoost classifier varied across the PMA groups (Table 2). The optimal hyperparameter configurations identified through grid search for the global XGBoost were consistent across the PMA groups (eTable 3, <http://links.lww.com/PR9/A340> in Supplementary Materials). The performance of the machine learning models (XGBoost, SVM, RF, LR, and CNN) in classifying the pre-lance epoch vs post-lance epoch in preterm infants was evaluated

Table 2
Important features identified by XGBoost.

PMA group	Features
Youngest	T7_FB1_spectral_flatness T8_FB1_amplitude_skew T7_FB2_spectral_flatness T7_FB2_spectral_entropy _FB3_connectivity_coh_freqmax T7_FB1_spectral_entropy F7_FB1_spectral_flatness F8_FB3_spectral_flatness
Middle	T8_FB2_spectral_flatness P8_FB1_spectral_flatness T8_FB2_spectral_entropy F7_FB2_spectral_relative_power T7_FB3_spectral_flatness T7_FB3_spectral_entropy
Oldest	T7_FB3_amplitude_skew T8_FB2_spectral_flatness P7_FB2_spectral_flatness O2_FB3_spectral_flatness O1_FB4_amplitude_skew O2_FB1_amplitude_kurtosis O2_FB1_amplitude_skew P7_FB2_spectral_entropy O2_FB3_spectral_entropy

The feature names follow the format: Channel_FrequencyBand_FeatureType. For example, "T8_FB2_spectral_flatness" indicates that the feature was extracted from the T8 channel. "FB2" corresponds to the frequency band from 3 to 8 Hz, and "spectral_flatness" is the type of feature extracted from the NEURAL toolbox over the one-second epoch. PMA, postmenstrual age.

across the 5 outer folds in the 3 PMA groups separately (Table 3).

3.1. Youngest postmenstrual age group (≤ 31 weeks, 6 days)

In the youngest group, XGBoost feature-importance weights ranked the top 4 predictors as F7_FB1_spectral_flatness (24.9%), T7_FB1_spectral_flatness (20.0%), T8_FB1_amplitude_skew (16.1%), and FB3_connectivity_coh_freqmax (12.5%).

The overall mean accuracy (ie, percentage of pre-lance and post-lance epochs correctly classified) ranged between 73% and 77%, with the SVM and RF models performing the best in this age group. The mean precision values ranged from 0.71 (XGBoost) to 0.82 (RF), indicating variability in correctly identifying true post-lance epochs from those predicted as post-lance. The average recall, which measures the model's ability to correctly identify all actual post-lance epochs, ranged from 0.71 (RF) to 0.79 (SVM). The highest specificity (ie, proportion of pre-lance epochs that are correctly identified) was 0.83 for the RF model. The SVM and CNN models each achieved an average AUC of 0.82, reflecting how well they distinguish post-lance from pre-lance epochs. This was followed by the RF and LR models with a mean AUC of 0.80. The XGBoost model showed slightly lower performance with a mean AUC of 0.77.

3.2. Middle postmenstrual age group (32–33 weeks, 6 days)

According to the XGBoost feature-importance weights, the top 4 predictors were: T8_FB2_spectral_flatness (38.5%), P8_FB1_spectral_flatness (20%), T8_FB2_spectral_entropy (16.3%), and F7_FB2_spectral_relative_power (12%).

The overall mean accuracy ranged from 58% to 70%, with the LR model achieving the highest mean accuracy of 70%. The mean precision achieved by the models varied from 0.47 (CNN) to 0.69 (LR). The mean recall varied, with some models achieving 0.55 (XGBoost), while the RF and LR models achieved a true positive rate of 0.75. Specificity ranged from 0.55 (XGBoost) to

0.65 (LR). The LR model achieved the highest mean AUC value of 0.76. The remaining models' average AUC ranged from 0.66 to 0.73.

3.3. Oldest postmenstrual age group (≥ 34 weeks)

According to XGBoost feature-importance weights, the top 4 predictors were T7_FB3_amplitude_skew (17.0%), T8_FB2_spectral_flatness (15.9%), P7_FB2_spectral_flatness (15.3%), and O2_FB3_spectral_flatness (14.8%).

The overall mean accuracy ranged from 70% to 82%, with the LR achieving the highest accuracy of 82%. The mean precision values ranged from 0.71 (RF) to 0.87 (LR). Similarly, the mean recall was also within a similar range of 0.68 (RF) to 0.88 (CNN). The mean specificity ranged from 0.64 (CNN) to 0.88 (LR). The LR model achieved the highest mean AUC value of 0.90, followed by the SVM with an average AUC of 0.88. The CNN and XGBoost models had a slightly lower mean AUC of 0.86, while the RF model had an AUC of 0.83.

4. Discussion

In this study, machine learning classifiers demonstrated the ability to distinguish one-second EEG epochs immediately before and after a clinically required heel lance by leveraging multiple EEG channels and a comprehensive array of features with data obtained at 2 separate clinical sites varying on sucrose administration. Performance varied systematically across developmental stages. Predictive accuracy and discrimination (as summarized by AUC) were highest in the most premature (≤ 31 weeks, 6 days PMA) and near-term (≥ 34 weeks PMA) groups, and showed a clear dip in the mid-prematurity window (32–33 weeks, 6 days PMA). This U-shaped pattern aligns with neurodevelopmental variability. Infants at different PMA stages exhibit distinct EEG patterns, reflecting the ongoing development of their nervous systems.^{24,36} More specifically, between approximately 29- and 35-week postconception, the preterm cortex undergoes profound circuit

Table 3
Summary of metrics of binary classification across preterm infants postmenstrual age groups.

PMA group	Model	Accuracy	Precision	Recall	Specificity	F1-score	AUC
Youngest	XGBoost	0.73 $\pm$ 0.13	0.71 $\pm$ 0.11	0.75 $\pm$ 0.20	0.71 $\pm$ 0.09	0.72 $\pm$ 0.15	0.77 $\pm$ 0.15
	SVM	0.77 $\pm$ 0.12	0.77 $\pm$ 0.15	0.79 $\pm$ 0.18	0.75 $\pm$ 0.16	0.77 $\pm$ 0.13	0.82 $\pm$ 0.10
	RF	0.77 $\pm$ 0.14	0.82 $\pm$ 0.18	0.71 $\pm$ 0.19	0.83 $\pm$ 0.17	0.75 $\pm$ 0.15	0.80 $\pm$ 0.15
	LR	0.76 $\pm$ 0.12	0.76 $\pm$ 0.11	0.78 $\pm$ 0.16	0.75 $\pm$ 0.14	0.76 $\pm$ 0.12	0.80 $\pm$ 0.10
	CNN	0.76 $\pm$ 0.09	0.77 $\pm$ 0.06	0.75 $\pm$ 0.16	0.78 $\pm$ 0.06	0.75 $\pm$ 0.10	0.82 $\pm$ 0.07
Middle	XGBoost	0.58 $\pm$ 0.14	0.58 $\pm$ 0.14	0.55 $\pm$ 0.21	0.60 $\pm$ 0.22	0.55 $\pm$ 0.17	0.73 $\pm$ 0.14
	SVM	0.63 $\pm$ 0.32	0.57 $\pm$ 0.34	0.70 $\pm$ 0.41	0.55 $\pm$ 0.27	0.63 $\pm$ 0.37	0.66 $\pm$ 0.31
	RF	0.65 $\pm$ 0.24	0.62 $\pm$ 0.23	0.75 $\pm$ 0.31	0.55 $\pm$ 0.27	0.67 $\pm$ 0.25	0.73 $\pm$ 0.23
	LR	0.70 $\pm$ 0.21	0.69 $\pm$ 0.25	0.75 $\pm$ 0.31	0.65 $\pm$ 0.29	0.70 $\pm$ 0.24	0.76 $\pm$ 0.24
	CNN	0.58 $\pm$ 0.11	0.47 $\pm$ 0.27	0.60 $\pm$ 0.38	0.55 $\pm$ 0.33	0.51 $\pm$ 0.30	0.65 $\pm$ 0.22
Oldest	XGBoost	0.74 $\pm$ 0.13	0.79 $\pm$ 0.20	0.72 $\pm$ 0.23	0.76 $\pm$ 0.26	0.73 $\pm$ 0.15	0.86 $\pm$ 0.16
	SVM	0.72 $\pm$ 0.18	0.76 $\pm$ 0.23	0.72 $\pm$ 0.23	0.72 $\pm$ 0.30	0.72 $\pm$ 0.19	0.88 $\pm$ 0.17
	RF	0.70 $\pm$ 0.10	0.71 $\pm$ 0.08	0.68 $\pm$ 0.23	0.72 $\pm$ 0.11	0.68 $\pm$ 0.14	0.83 $\pm$ 0.19
	LR	0.82 $\pm$ 0.16	0.87 $\pm$ 0.18	0.76 $\pm$ 0.22	0.88 $\pm$ 0.18	0.80 $\pm$ 0.19	0.90 $\pm$ 0.12
	CNN	0.76 $\pm$ 0.11	0.75 $\pm$ 0.16	0.88 $\pm$ 0.18	0.64 $\pm$ 0.30	0.79 $\pm$ 0.09	0.86 $\pm$ 0.13

Mean $\pm$ SD of classification metrics across 5 outer test-folds (nested cross-validation). Models examined: XGBoost, support vector machines (SVM), random forest (RF), logistic regression (LR), and convolutional neural networks (CNN). Youngest group: infants with a postmenstrual age (PMA) of 31 weeks, 6 days, or less at the time of the study (n = 27). Middle group: infants with a PMA between 32 weeks and 33 weeks, 6 days at the time of the study (n = 20). Oldest group: infants with a PMA of 34 weeks or greater at the time of the study (n = 25). AUC, area under the receiver operating characteristic curve.

refinement.⁶³ A 2010 review by Kostović and Judaš demonstrated that before 32 weeks, most infants exhibit relatively homogeneous, subplate-mediated burst activity, whereas by 34 weeks and beyond, the thalamocortical network has matured into a more stable, sensory-driven response. During the transitional 32 to 34 week window, however, some infants retain immature subplate bursts while others already display mature, stereotyped evoked potentials.²⁴ This developmental heterogeneity produces variable EEG signatures that likely account for the reduced model performance observed in the mid-PMA cohort.

It should be noted that our middle cohort was also the smallest ($n = 20$) compared with the youngest ($n = 27$) and oldest ($n = 25$) groups. In addition to any physiological variability due to neurodevelopmental considerations, fewer subjects translate into less training data and greater variance in model estimates.

These findings suggest that “one-size-fits-all” pain classifiers may struggle most at precisely the ages when the human cortex is undergoing its most dramatic reorganization.¹² Age-specific models implemented in ML tools are likely necessary to capture the nuances of developmental differences within the first weeks of life. Furthermore, the varying performances of the different models across PMA groups highlight the need for a tailored approach, potentially combining multiple classification models to leverage their respective strengths.

4.1. Importance of features

Feature weight importance indicates how often and how heavily the tree algorithm used each feature in its internal splits. Our machine-learning models consistently selected a small set of EEG-derived features as the strongest predictors of post-lance vs pre-lance cortical responses across all age groups.

Notably, sucrose administration was neither retained nor influential in the final feature set, demonstrating that it was not a statistically significant predictor of pre-lance vs post-lance epoch classification. Moreover, low-frequency (0.5–8 Hz bands) spectral flatness and amplitude skewness were primary drivers for the classification. However, it is important to emphasize that feature-importance scores describe statistical associations, not causal mechanisms. That is, there is limited interpretation possible as these do not represent biomarkers of the brain processing noxious stimulus and only indicate which inputs the model leaned on most heavily to reduce error.

4.2. Model performance in relation to previous literature

In comparison with previous literature,^{58,59} the EEG-only RF model achieved a 0.64 accuracy and 0.75 AUC when distinguishing pre-lance and post-lance epochs in their youngest group, below the 0.73 to 0.77 accuracy and 0.77 to 0.82 AUC achieved by our models. In the middle group, prior models achieved an AUC of 0.89, which exceeds the 0.66 to 0.76 AUC observed in this study, highlighting a challenge in this PMA range. Van der Vaart and colleagues demonstrated 0.81 accuracy and 0.90 AUC in a ≥ 34 weeks PMA test set, the results that align closely with the results achieved in oldest group in this study.⁵⁹ Support vector machines and XGBoost also demonstrated strong discrimination. Overall, the current multisite, multiple EEG channels, and multifeature pipeline extends prior work by improving discrimination in the youngest infants and matching performance in the oldest infants, while the middle PMA group remains a key target for future model refinement. Finally, our approach took a more autonomous approach in that no data were preprocessed or analyzed using human judgment.

4.3. Clinical implications

Although previous literature has emphasized accuracy and discrimination, clinically acceptable thresholds for evaluation metrics depend on the context and the specific medical decision being made.²¹ In the preterm infant pain context, ensuring that ongoing pain-related cortical responses are identified is paramount, making recall (ie, the proportion of true post-lance epochs correctly flagged) a key metric to consider. Across our five-fold outer validation, average recall reached 0.79 in the youngest PMA group, 0.75 in the middle group, and 0.88 in the oldest group, meaning fewer than 25% of pain-related epochs were undetected. Clinically, this high recall reduces the risk that a true nociceptive response goes unnoticed, which is critical when aiming to avoid under-treatment of neonatal pain in the NICU.

4.4. Limitations and future directions

One potential limitation of this study is that the administration of sucrose before the heel lance was site-dependent. Although some studies have suggested that sucrose administration can modulate behavioural and physiological pain markers,⁵⁵ it was neither highly discriminative nor retained in our final feature set, indicating it did not drive model predictions. This may reinforce the idea that, while sucrose can reduce overt pain behaviours, it does not alter noxious-evoked cortical activity in neonates.⁵⁴ In general, the variability in sucrose administration could strengthen the model by introducing heterogeneity, making it more robust and generalizable across different clinical practices. However, future studies with larger samples should continue to monitor and explicitly model sucrose effects.

Furthermore, our study focused exclusively on clinically required heel lance procedures, which limits the generalizability of our findings to other potentially painful procedures, or to pain lasting longer than 1 second such as postoperative pain. However, the preliminary classification models developed provide a solid foundation for distinguishing brain activity during skin-breaking procedures from baseline activity. Finally, our future work will also incorporate multimodal assessments, including heart rate, respiration rate, oxygen saturation, and facial behavioral coding to improve performance.

5. Conclusion

The results from this study demonstrate the feasibility of using machine learning to discriminate preterm infants' cortical activity after pain-related stimulus from nonpain-related activity using EEG. By leveraging advanced machine learning techniques, using multiple EEG channels and ensuring a more automated and robust feature extraction process, we address several limitations found in existing studies. This is a step toward autonomous real-time pain assessments in NICUs, improving the accuracy and consistency of pain management for the most vulnerable patients.

Disclosures

There are no conflicts of interests to declare.

Acknowledgements

Nonauthor collaborators: The authors acknowledge the full team of investigators and coinvestigators who helped on this project: Lesley Johannsson, Maria Pureza Laudiano-Dray, Neelum Mistry, Laura Jones, Mohamed Rupawala, Oana Bucsea,

Ilana Shiff, Haleh Hashemi, and Estreya Cohen. The authors acknowledge Affan Jilani for technical help related to analysis.

Funding: The authors acknowledge the Natural Sciences and Engineering Research Council of Canada (NSERC; Funding Reference Number: 538853-2019), CIHR (Collaborative Health Research Project; Funding Reference Number CPG-163988), Social Sciences and Humanities Research Council of Canada (SSHRC; Funding Reference Number: 1001-2019-0004; CGS-M Award to L.H.), and the York Research Chair Program for financial support of this project. The authors acknowledge all research and clinical staff as well as the infants and families who have made this project possible.

Data availability: Data will be made available by authors upon reasonable request.

Supplemental digital content

Supplemental digital content associated with this article can be found online at <http://links.lww.com/PR9/A340>.

Article history:

Received 7 October 2024

Received in revised form 26 May 2025

Accepted 30 June 2025

Available online 16 September 2025

References

- [1] Ansari AH, Cherian PJ, Caicedo Dorado A, Jansen K, Dereymaeker A, De Wispelaere L, Dielman C, Vervisch J, Govaert P, De Vos M, Naulaers G, Huffel SV. Weighted performance metrics for automatic neonatal seizure detection using multiscored EEG data. *IEEE J Biomed Health Inform* 2018;22:1114–23.
- [2] Ansari AH, Pillay K, Dereymaeker A, Jansen K, Van Huffel S, Naulaers G, De Vos M. A deep shared multi-scale inception network enables accurate neonatal quiet sleep detection with limited EEG channels. *IEEE J Biomed Health Inform* 2022;26:1023–33.
- [3] Baud O, Arzounian D, Bourel-Ponchel E. Continuous monitoring of neonatal cortical activity: a major step forward. *Cell Rep Med* 2022;3:100864.
- [4] Benoit B, Martin-Misener R, Newman A, Latimer M, Campbell-Yeo M. Neurophysiological assessment of acute pain in infants: a scoping review of research methods. *Acta Paediatr* 2017;106:1053–66.
- [5] Boyle EM, Bradshaw J, Blake KI. Persistent pain in neonates: challenges in assessment without the aid of a clinical tool. *Acta Paediatr* 2018;107:63–7.
- [6] Brummelte S, Grunau RE, Chau V, Poskitt KJ, Brant R, Vinall J, Gover A, Synnes AR, Miller SP. Procedural pain and brain development in premature newborns. *Ann Neurol* 2012;71:385–96.
- [7] Carlini LP, Coutin GdAS, Ferreira LA, Soares JdCA, Silva GVT, Heiderich TM, Balda RdCX, Barros MCDM, Guinsburg R, Thomaz CE. Human vs machine towards neonatal pain assessment: a comprehensive analysis of the facial features extracted by health professionals, parents, and convolutional neural networks. *Artif Intelligence Med* 2024;147:102724.
- [8] Chawanpaiboon S, Vogel JP, Moller AB, Lumbiganon P, Petzold M, Hogan D, Landoulsi S, Jampathong N, Kongwattanakul K, Laopaiboon M, Lewis C, Rattanakankokchai S, Teng DN, Thinkhamrop J, Watananirun K, Zhang J, Zhou W, Gülmezoglu AM. Global, regional, and national estimates of levels of preterm birth in 2014: a systematic review and modelling analysis. *Lancet Glob Health* 2019;7:e37–46.
- [9] Cong X, McGrath JM, Cusson RM, Zhang D. Pain assessment and measurement in neonates: an updated review. *Adv Neonatal Care* 2013;13:379–95.
- [10] Dilella R, Raviglione F, Cantalupo G, Cordelli DM, De Liso P, Di Capua M, Falsaperla R, Ferrari F, Fumagalli M, Lori S, Suppiej A, Tadini L, Dalla Bernardina B, Mastrangelo M, Pisani F. Consensus protocol for EEG and amplitude-integrated EEG assessment and monitoring in neonates. *Clin Neurophysiol* 2021;132:886–903.
- [11] Duerden EG, Grunau RE, Guo T, Foong J, Pearson A, Au-Young S, Lavie R, Chakravarty MM, Chau V, Synnes A, Miller SP. Early procedural pain is associated with regionally-specific alterations in thalamic development in preterm neonates. *J Neurosci* 2018;38:878–86.
- [12] Garcia KE, Robinson EC, Alexopoulos D, Dierker DL, Glasser MF, Coalson TS, Ortinau CM, Rueckert D, Taber LA, Van Essen DC, Rogers CE, Smyser CD, Bayly PV. Dynamic patterns of cortical expansion during folding of the preterm human brain. *Proc Natl Acad Sci U S A* 2018;115:3156–61.
- [13] Gkikas S, Tsiknakis M. Automatic assessment of pain based on deep learning methods: a systematic review. *Computer Methods Programs Biomed* 2023;231:107365.
- [14] Grunau RE, Vinall Miller J, Chau CM. Long-term effects of pain in infants. *Oxford Textbook of Pediatric Pain*, Vol. 38, 2021. p. 38–46.
- [15] Gschwandtner L, Hartmann M, Oberdorfer L, et al. Deep learning for estimation of functional brain maturation from EEG of premature neonates. 42nd Annu Int Conf IEEE Eng Med Biol Soc 2020;2020:104–7.
- [16] Hartley C, Baxter L, Moultrie F, Purdy R, Bhatt A, Rogers R, Patel C, Adams E, Slater R. Predicting severity of adverse cardiorespiratory effects of morphine in premature infants: a post hoc analysis of procedural pain in premature infants trial data. *Br J Anaesth* 2021;126:e133–5.
- [17] Hartley C, Duff EP, Green G, Mellado GS, Worley A, Rogers R, Slater R. Nociceptive brain activity as a measure of analgesic efficacy in infants. *Sci Transl Med* 2017;9:eaah6122.
- [18] Hatfield LA, Ely EA. Measurement of acute pain in infants: a review of behavioral and physiological variables. *Biol Res Nurs* 2015;17:100–11.
- [19] Hatfield LA, Murphy N, Karp K, Polomano RC. A systematic review of behavioral and environmental interventions for procedural pain management in preterm infants. *J Pediatr Nurs* 2019;44:22–30.
- [20] Heiderich TM, Carlini LP, Buzuti LF, Balda RdC, Barros MC, Guinsburg R, Thomaz CE. Face-based automatic pain assessment: challenges and perspectives in neonatal intensive care units. *J Pediatr* 2023;99:546–60.
- [21] Hicks SA, Strümke I, Thambawita V, Hammou M, Riegler MA, Halvorsen P, Parasa S. On evaluation metrics for medical applications of artificial intelligence. *Sci Rep* 2022;12:5979.
- [22] Hughes JD, Chivers P, Hoti K. The clinical suitability of an artificial intelligence-enabled pain assessment tool for use in infants: feasibility and usability evaluation study. *J Med Int Res* 2023;25:e41992.
- [23] Jones L, Fabrizi L, Laudiano-Dray M, Whitehead K, Meek J, Verriotti M, Fitzgerald M. Nociceptive cortical activity is dissociated from nociceptive behavior in newborn human infants under stress. *Curr Biol* 2017;27:3846–51.e3.
- [24] Kostović I, Judas M. The development of the subplate and thalamocortical connections in the human foetal brain. *Acta Paediatr* 2010;99:1119–27.
- [25] Llerena A, Tran K, Choudhary D, Hausmann J, Goldgof D, Sun Y, Prescott SM. Neonatal pain assessment: do we have the right tools? *Front Pediatr* 2022;10:1022751.
- [26] Lopez KL, Monachino AD, Morales S, Leach SC, Bowers ME, Gabard-Durnam LJ. HAPPILEE: HAPPE in low electrode electroencephalography, a standardized pre-processing software for lower density recordings. *Neuroimage*. 2022;260:119390.
- [27] Maimon N, Grunau RE, Cepeda IL, Friger M, Selnovik L, Gilat S, Shany E. Electroencephalographic activity in response to procedural pain in preterm infants born at 28 and 33 weeks gestational age. *Clin J Pain* 2013;29:1044–9.
- [28] Manworren RCB, Homer S, Joseph R, Dadar P, Kaduwela N. Performance evaluation of a supervised machine learning pain classification model developed by neonatal nurses. *Adv Neonatal Care* 2024;24:301–10.
- [29] Marchant S, van der Vaart M, Pillay K, Baxter L, Bhatt A, Fitzgibbon S, Hartley C, Slater R. A machine learning artefact detection method for single-channel infant event-related potential studies. *J Neural Eng* 2024;21:046021.
- [30] Mari T, Henderson J, Maden M, Nevitt S, Duarte R, Fallon N. Systematic review of the effectiveness of machine learning algorithms for classifying pain intensity, phenotype or treatment outcomes using electroencephalogram data. *J Pain* 2022;23:349–69.
- [31] The MathWorks Inc. MATLAB version 23.2 (R2023b). Natick: The MathWorks Inc.; 2023.
- [32] Matsangidou M, Liampas A, Pittara M, Pattichi CS, Zis P. Machine learning in pain medicine: an up-to-date systematic review. *Pain Ther* 2021;10:1067–84.
- [33] Montazeri S, Airaksinen M, Nevalainen P, Marchi V, Hellström-Westas L, Stevenson NJ, Vanhatalo S. An automated bedside measure for monitoring neonatal cortical activity: a supervised deep learning-based electroencephalogram classifier with external cohort validation. *Lancet Digital Health* 2022;4:e884–92.
- [34] Murphy BM, Goulding RM, O'Toole JM. Detection of transient bursts in the EEG of preterm infants using time-frequency distributions and machine learning. *Annu Int Conf IEEE Eng Med Biol Soc* 2020;2020:1023–6.
- [35] Murphy K, Stevenson NJ, Goulding RM, Lloyd RO, Korotchnikova I, Livingstone V, Boylan GB. Automated analysis of multi-channel EEG in preterm infants. *Clin Neurophysiol* 2015;126:1692–702.

- [36] Myers MM, Grieve PG, Izraelit A, Fifer W, Isler J, Darnall R, Stark R. Developmental profiles of infant EEG: Overlap with transient cortical circuits. *Clin Neurophysiol* 2012;123:1502–11.
- [37] Nath N, Beltrano W, Haynes L, Dewey D, Bray S. Long-term effects of preterm birth on children's brain structure: an analysis of the adolescent brain cognitive development (ABCD) study. *eNeuro* 2023;10: ENEURO.0196-22.2023.
- [38] Navarro X, Porée F, Kuchenbuch M, Chavez M, Beuchée A, Carrault G. Multi-feature classifiers for burst detection in single EEG channels from preterm infants. *J Neural Eng* 2017;14:046015.
- [39] Ohuma EO, Moller AB, Bradley E, Chakwera S, Hussain-Alkhateeb L, Lewin A, Okwaraji YB, Mahanani WR, Johansson EW, Lavin T, Fernandez DE, Dominguez GG, de Costa A, Cresswell JA, Krasevec J, Lawn JE, Blencowe H, Requejo J, Moran AC. National, regional, and global estimates of preterm birth in 2020, with trends from 2010: a systematic analysis. *Lancet* 2023;402:1261–71.
- [40] O'Shea A, Ahmed R, Lightbody G, Pavlidis E, Lloyd R, Pisani F, Mamane W, Mathieson S, Boylan G, Temko A. Deep learning for EEG seizure detection in preterm infants. *Int J Neural Syst* 2021;31:2150008.
- [41] O'Toole JM, Boylan GB. Quantitative preterm EEG analysis: the need for caution in using modern data science techniques. *Front Pediatr* 2019;7: 174.
- [42] O'Toole JM, Boylan GB. NEURAL: quantitative features for newborn EEG using Matlab. *arXiv preprint*. 2017. doi:10.48550/arXiv.1704.05694.
- [43] Palmu K, Wikström S, Hippeläinen E, Boylan G, Hellström-Westas L, Vanhatalo S. Detection of "EEG bursts" in the early preterm EEG: visual vs. automated detection. *Clin Neurophysiol* 2010;121:1015–22.
- [44] Pavel AM, Rennie JM, de Vries LS, Blennow M, Foran A, Shah DK, Pressler RM, Kapellou O, Dempsey EM, Mathieson SR, Pavlidis E, van Huffelen AC, Livingstone V, Toet MC, Weeke LC, Finder M, Mitra S, Murray DM, Mamane WP, Boylan GB. A machine-learning algorithm for neonatal seizure recognition: a multicentre, randomised, controlled trial. *Lancet Child Adolesc Health* 2020;4:740–9.
- [45] Pillai Riddell RR, Jasim S, Hamwi L. Out of the mouth of babes: a lot about pain has nothing to do with pain. *PAIN* 2022;163:S117–25.
- [46] Racine N, Chow C, Hamwi L, Bucsea O, Cheng C, Du H, Fabrizio L, Jasim S, Johannsson L, Jones L, Laudiano-Dray MP, Meek J, Mistry N, Shah V, Stedman I, Wang X, Riddell RP. Health care professionals' and parents' perspectives on the use of AI for pain monitoring in the neonatal intensive care unit: multisite qualitative study. *JMIR AI* 2024;3:e51535.
- [47] Roué JM, Rioualen S, Gendras J, Misery L, Gouillou M, Sizun J. Multimodal pain assessment: are near-infrared spectroscopy, skin conductance, salivary cortisol, physiologic parameters, and neonatal facial coding system interrelated during venepuncture in healthy, term neonates? *J Pain Res* 2018;11:2257–67.
- [48] Routier L, Querne L, Ghostine-Ramadan G, Boulesteix J, Graïc S, Mory S, Wallois F, Bourel-Ponchel E. Predicting the neurodevelopmental outcome in extremely preterm newborns using a multimodal prognostic model including brain function information. *JAMA Netw Open* 2023;6: e231590.
- [49] Rupawala M, Bucsea O, Laudiano-Dray MP, Whitehead K, Meek J, Fitzgerald M, Olhede S, Jones L, Fabrizio L. A developmental shift in habituation to pain in human neonates. *Curr Biol* 2023;33:1397–406.e5.
- [50] Salekin MS, Zamzmi G, Goldgof D, Kasturi R, Ho T, Sun Y. Multimodal spatio-temporal deep learning approach for neonatal postoperative pain assessment. *Comput Biol Med* 2021;129:104150.
- [51] San-Segundo R, Gil-Martín M, D'Haro-Enríquez LF, Pardo JM. Classification of epileptic EEG recordings using signal transforms and convolutional neural networks. *Comput Biol Med* 2019;109:148–58.
- [52] Shiff I, Bucsea O, Pillai Riddell R. Psychosocial and neurobiological vulnerabilities of the hospitalized preterm infant and relevant non-pharmacological pain mitigation strategies. *Front Pediatr* 2021;9:568755.
- [53] Shiroshita Y, Kirimoto H, Ozawa M, Watanabe T, Uematsu H, Yunoki K, Sobue I. Can event-related potentials evoked by heel lance assess pain processing in neonates? A systematic review. *Children* 2021;8:58.
- [54] Slater R, Fabrizio L, Worley A, Meek J, Boyd S, Fitzgerald M. Premature infants display increased noxious-evoked neuronal activity in the brain compared to healthy age-matched term-born infants. *Neuroimage* 2010; 52:583–9.
- [55] Stevens B, Yamada J, Ohlsson A, Haliburton S, Shorkey A. Sucrose for analgesia in newborn infants undergoing painful procedures. *Cochrane database Syst Rev* 2016;7:CD001069.
- [56] Stevenson NJ, Tataranno ML, Kaminska A, Pavlidis E, Clancy RR, Griesmaier E, Roberts JA, Klebermass-Schrehof K, Vanhatalo S. Reliability and accuracy of EEG interpretation for estimating age in preterm infants. *Ann Clin Translat Neurol* 2020;7:1564–73.
- [57] Stoye DQ, Boardman JP, Osmond C, Sullivan G, Lamb G, Black GS, Homer NZM, Nelson N, Theodorsson E, Mörelus E, Reynolds RM. Saliva cortisol diurnal variation and stress responses in term and preterm infants. *Arch Dis Child Fetal Neonatal Ed* 2022;107:558–64.
- [58] van der Vaart M, Duff E, Raafat N, Rogers R, Hartley C, Slater R. Multimodal pain assessment improves discrimination between noxious and non-noxious stimuli in infants. *Paediatric Neonatal Pain* 2019;1:21–30.
- [59] van der Vaart M, Hartley C, Baxter L, Mellado GS, Andritsou F, Cobo MM, Fry RE, Adams E, Fitzgibbon S, Slater R. Premature infants display discriminable behavioral, physiological, and brain responses to noxious and non-noxious stimuli. *Cereb Cortex* 2022;32:3799–815.
- [60] Verriotis M, Fabrizio L, Lee A, Cooper RJ, Fitzgerald M, Meek J. Mapping cortical responses to somatosensory stimuli in human infants with simultaneous near-infrared spectroscopy and event-related potential recording. *eNeuro* 2016;3:ENEURO.0026-16.2016.
- [61] Vinall J, Grunau RE. Impact of repeated procedural pain-related stress in infants born very preterm. *Pediatr Res* 2014;75:584–7.
- [62] Williams MD, Lascelles BD. Early neonatal pain—a review of clinical and experimental implications on painful conditions later in life. *Front Pediatr* 2020;8:30.
- [63] Zamzmi G, Pai CY, Goldgof D, Kasturi R, Ashmeade T, Sun Y. An approach for automated multimodal analysis of infants' pain. In: 2016 23rd International Conference on Pattern Recognition (ICPR), 2016. p. 4148–53. doi: 10.1109/ICPR.2016.7900284
- [64] Zamzmi G, Goldgof D, Kasturi R, Sun Y, Ashmeade T. Machine-based multimodal pain assessment tool for infants: A review, 2019. Available at: <http://arxiv.org/abs/1607.00331>. Accessed September 28, 2023.
- [65] Zhang M, Zhu L, Lin SY, Herr K, Chi CL, Demir I, Dunn Lopez K, Chi NC. Using artificial intelligence to improve pain assessment and pain management: a scoping review. *J Am Med Inform Assoc* 2023;30: 570–87.